



PIYUSH MAJI

Master of Science in Physics  
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## EDUCATION

Indian Institute of Technology Hyderabad

Expected July 2027

M.Sc. in Physics CGPA: 7.5

Hooghly Mohsin College (1836), West Bengal

August 2024

B.Sc. in Physics CGPA: 7.28

## RESEARCH

Graph-based Full Event Interpretation (GraFEI) | Belle II Experiment

Ongoing

Python • PyTorch Geometric • GNNs • basf2 | Advisor: Prof. Sourav Sandilya, Belle II Collaboration

- Addressed the challenge of reconstructing complex  $B$ -meson decay topologies without predefined decay-chain assumptions by **developing** an end-to-end **Graph Neural Network (GNN)** pipeline that learns directly from detector-level particle observations.
- Engineered** fully connected graph representations using particle kinematics and detector observables (DOCA,  $\cos \theta$ , momentum, charge), and **implemented** a multi-task GNN in **PyTorch Geometric** to jointly predict particle identities and **hierarchical decay structures (LCAS)**.
- Validated** the model within the Belle II basf2 framework using **PerfectLCA**, **PerfectMasses**, and **PerfectEvent** metrics, contributing to ML-driven event reconstruction for rare **high-energy physics** analyses.

## PROJECTS

GNN-FraudNet | Graph-based Financial Fraud Detection

Self-Initiated • Completed

Python • PyTorch Geometric • FastAPI • XGBoost • Docker [GitHub](#)

- Addressed the challenge of detecting fraudulent Bitcoin transactions, where traditional ML models ignore transaction relationships, by **developing** a **Graph Neural Network** pipeline that leverages graph connectivity for node classification.
- Engineered** a graph learning workflow on the **Elliptic Bitcoin Dataset (203K+ transactions, 234K+ edges)** using **GraphSAGE** and **Graph Attention Networks (GAT)**, with preprocessing, class balancing, and explainability through **SHAP** and **GNNExplainer**.
- Built** a production-ready inference pipeline by exposing fraud predictions through a **FastAPI** REST API and containerizing the application with **Docker** for scalable deployment.

Retail Demand Forecasting & Inventory Optimization

Self-Initiated • Ongoing

Python • LightGBM • Time Series Forecasting [GitHub](#)

- Solved** the problem of retail demand forecasting by building a scalable **SKU-level sales prediction pipeline** using **LightGBM**, enabling accurate inventory planning across multiple stores.
- Designed** advanced time-series features including lag, rolling-window, calendar, promotion, and holiday variables, and benchmarked the model against **SARIMA** and **Prophet** baselines using time-aware cross-validation.
- Implemented hierarchical forecast reconciliation** to maintain consistency across SKU, store, and regional forecasts, improving usability for supply chain decision-making.

Spam Classification from Scratch

Self-Initiated • Completed

Python • NumPy • Machine Learning [GitHub](#)

- Built** a machine learning pipeline from scratch to detect spam SMS messages, implementing **Multinomial Naive Bayes** and **Logistic Regression** without using scikit-learn or other ML libraries.
- Implemented** custom Bag-of-Words, **TF-IDF**, MLE, Laplace smoothing, and gradient descent, with end-to-end evaluation (**ROC**, **AUC**, **F1**) while ensuring data-leakage-free feature engineering.
- Achieved 94.9% F1-score** and **0.982 AUC** on the UCI SMS Spam Collection, demonstrating strong performance through threshold optimization and comparative model analysis.

## POSITIONS OF RESPONSIBILITY

Placement Coordinator, Office of Career Services, IIT Hyderabad

May 2026 – Present

Manage end-to-end campus recruitment drives and liaise with recruiters; **hosted Japan Career Day 2026** and **Career Fair 2026**, IIT Hyderabad.

Inbound & Logistics, International Relations Cell, IIT Hyderabad

Sep 2025 – Present

Coordinate orientations, cultural programs, and academic tours for **exchange students from Japan**, **international delegates**, and **company officials**.

Core Member, Extra Mural Lectures (EML), IIT Hyderabad

May 2026 – Present

Lead visual design and event branding for EML; **managed an INR 100K budget**; hosted **NASA Axiom Mission Pilot Subhanshu Shukla** for a campus lecture.

## SKILLS

**Languages & Tools:** Python, SQL, Git, NumPy, Pandas, PyTorch, PyTorch Geometric, LightGBM, basf2

**Domain:** Machine Learning, Graph Neural Networks, A/B Testing & Experimentation, Time Series Forecasting, Data Science, Data Structures & Algorithms

**Coding Practice:** LeetCode, GeeksforGeeks, Striver A2Z, Codeforces

## RELEVANT COURSES

Machine Learning for Physical Sciences • Machine Learning for High Energy Physics • Statistical Mechanics • Linear Algebra • Computational Physics • Data Science Analysis

## CERTIFICATIONS

Machine Learning Specialization – DeepLearning.AI / Coursera

## COMPETITIONS

Accenture Innovation Challenge 2026

Selected Finalist

Designed a **GNN-based digital twin** for vehicle assembly line optimization, modeling stations and workflow dependencies as graph structures to predict bottlenecks and throughput issues, applying graph representation learning techniques from concurrent Belle II / GNN-FraudNet research.